

$$\mathcal{H} = \frac{J_H}{M'} \sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j + \frac{\kappa}{(M')^2} \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \quad (1)$$

transformations:

$$\text{Eric : } \vec{S}_i = \sum_{\alpha\beta} f_{i\alpha}^\dagger \frac{\vec{\sigma}}{2} f_{i\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{1}{4} - \frac{1}{2} \sum_{\alpha\beta} f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} \quad (2)$$

$$\text{Pedro : } S_i^{\beta\alpha} = f_{i\beta}^\dagger f_{i\alpha} - \frac{Q}{M} \delta_{\alpha\beta} \Rightarrow \vec{S}_i \cdot \vec{S}_j = \frac{M}{4} - \sum_{\alpha\beta} f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} \quad (3)$$

difference of factor 2 for $M = 2$, which is my case. (Also, for me $M' = 1$ while Pedro used $M' = M$.)
mean field Hamiltonians:

$$\text{Eric : } \mathcal{H} \approx -J \sum_{\langle ij \rangle} \left[(1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c.} \right. \quad (4)$$

$$\left. + 2(-1 - \kappa/2 + 6\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \right] \quad (5)$$

$$\text{Pedro : } \mathcal{H} \approx -J \sum_{\langle ij \rangle} \left[2 + \boxed{2\kappa} - 16\kappa |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger (f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow}) + \text{h.c.} \right. \quad (6)$$

$$\left. + 2(-2 \boxed{-2\kappa} + 24\kappa |\langle \chi_{ij} \rangle|^2) |\langle \chi_{ij} \rangle|^2 \right] \quad (7)$$

Every term without a κ (coming from the bilinear exchange term) is off by a factor of 2. Every term with a κ (coming from the biquadratic exchange term) is off by a factor of 4. Hence, the only problem is the factor 2 in the fermionic representation. The boxed terms don't appear in Pedro's notes as they go like M^{-1} and are therefore irrelevant in the large M limit.